Double Number Line Ratio Explorer

Explore & Reflect

Name _____ Date _____ Period _____

Open **Double Number Line Ratio Explorer** in your unit's Interactive Math section on Mr. K's Classroom (or use your teacher's link). Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3. Do Parts 4 and 5 on your own.

1 Predict before you touch the tool

1. A recipe uses 3 cups of juice concentrate for every 5 cups of water. Will 5 cups of concentrate with 7 cups of water taste the same? Explain.

2. Guess: how much water goes with exactly 1 cup of concentrate?

2 Explore use the tool

1. Slide the marker to 9 cups of concentrate. How much water goes with it? What number multiplied both amounts?

2. Move the marker so the top line reads exactly 1. What does the bottom line show? What is this number called?

3. Now find how much concentrate goes with exactly 1 cup of water.

4. Type 5 and 7 into Test a pair. What does the tool say, and where is the point on the graph?

3 Record use the tool and fill in the table

Juice recipe, 3 : 5

Use the marker. The last column is "multiplied by" compared with 3 : 5.

Concentrate (cups)	Water (cups)	Water per cup of concentrate	Multiplied by
3	5	$\frac{5}{3}$	× 1
6			
	20		
$\frac{3}{2}$			
	1		

Road trip, 2 hours : 7 miles

Choose Road trip.

Hours	2	4	?1	?6
Miles	7			21

4 Explain what did you discover?

1. Equivalent ratios come from _____ both amounts by the same number.
2. The unit rate is the amount that goes with exactly _____. For equivalent ratios it is always _____.
3. On the graph, equivalent pairs lie on one straight line through _____.
4. A classmate says 4 : 6 and 6 : 8 are equivalent because you add 2 to each part. Explain why that is wrong. Use the unit rate or the double number line.

5 Apply a new situation, no tool

1. Apples cost $6 for 4 pounds. Find the price per pound and the cost of 10 pounds.
2. Which is the better buy: 3 notebooks for $4.50 or 5 notebooks for $7.00?
3. Fill in the blanks: 2 : 7 = 8 : ___ = ___ : 35
4. A car travels 150 miles in $2\frac{1}{2}$ hours. At the same rate, how far does it go in 4 hours?